

BENJAMIN MILLER

(925) 270-9677 | bem002@ucsd.edu | CA - Bay Area
LinkedIn | GitHub | Portfolio Website

EDUCATION

Computer Science B.S. - University of California, San Diego (UCSD) Sept. 2023 – June 2027
GPA: 3.8/4.0, Jacobs School of Engineering, UC San Diego Scholar's Society
Regent Scholar (Merit-based, top 1.5% of class)

Relevant Coursework: Advanced Data Structures and Algorithms (**C++**), Software Engineering, Operating Systems, Embedded Programming (**C**, **ARM Assembly**), Software Tools and Techniques (**Linux**, **Shell Scripts**), Data Science (**Machine Learning**, **Pandas**), Object Oriented Design (**Java**)

SKILLS

- **Programming:** Python, C, C++, Java, JavaScript, TypeScript, SQL
- **Web/App Development:** Flask, FastAPI, React, Node.js, REST APIs, Tailwind CSS, HTML, Vite, Express
- **Machine Learning:** NumPy, PyTorch, Pandas, scikit-learn, OpenCV
- **Tools & DevOps:** Linux, Git, Docker, CI/CD, Bash, Agile/Scrum, Virtual Machines

EXPERIENCE

Software Engineer Intern Jun. 2026 - Present, Sept. 2025 - Dec. 2025
Western Digital Irvine, CA

- Developed internal tool for checking server availability for integration testing via a **Python microservice** and a full stack application (**Flask**, **SQLite**, **JavaScript**), as seen by successful end to end live demonstrations to management.
- Designed and implemented **C++ APIs** for hard drives in an **Agile** development environment, validated through comprehensive unit and integration testing.
- Expanded and improved the **unit testing** suite, resulting in increased overall **QA validation** quality of the **CI/CD pipeline** and higher quality code.

Software Developer Intern Apr. 2025 - Aug. 2025
Center for Applied Internet Data Analysis San Diego, CA

- Enhanced maintainability and usability using **JavaScript** and **Python**, as demonstrated by successful contributions to the websites' codebase using **Git** for version control in large-scale site environments.
- Created useful automations using **Python** in a **Linux** environment resulting in a 75% decrease on average in manual labor required for workflows, such as content updates across the center's project sites.

Resident Advisor at COSMOS Jul. 2024 - Aug. 2024
Jacobs School of Engineering La Jolla, CA

- Provided mentorship for the Video Game Programming and AI Design group, as demonstrated by meaningful interactions and personalized advice to empower future engineers.

LEADERSHIP

Electronic Props Designer Mar. 2026 - Present
Star Wars Club at UCSD La Jolla, CA

- Designed embedded systems and **C++** firmware for props for film making and on-campus promotional events using a real time operating system and microcontrollers resulting in screen accurate prop recreations.

Software Engineering Lead & Vice President Jun. 2024 - Present
Themed Entertainment Association at UCSD La Jolla, CA

- Led development of software systems for themed attractions (**Python**, **C**, **C++**) which enhanced attraction operations that were attended by 100+ guests campuswide.

PROJECTS

Taro the Talking Bird - Interactive Robotic Figure Dec. 2025
C/C++, Multi-Threading, Advanced Linux Sound Architecture Demo Video | Project Page | Github

- Implemented a real-time **multithreaded system** using **C++** achieving audio processing with <10ms latency for synchronized motion control.
- Integrated local **AI models** for offline speech recognition and synthesis, processing **100+ live conversations** with 16K+ social media engagement.

Computer Vision Spell Detection Apr. 2026
Computer Vision, Python, Flask, Kotlin, MQTT Project Page | Github

- Built a distributed edge-AI system with a custom Android client and **backend** services for real-time gesture and speech-based event detection using computer vision and inference.
- Designed an event-driven architecture using MQTT to stream inference outputs from mobile devices to a centralized broker, enabling real-time processing and visualization via a **Python/Flask** dashboard.